Kiran Kishore Venkatesan

Agentic AI Engineer Intern | B.Tech – CSE (Artificial Intelligence) | **Amrita Vishwa Vidyapeetham, Coimbatore**
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Summary

AI engineer focused on building **robust, reliable AI systems that solve real-world problems**. Experienced in designing **autonomous multi-agent LLM systems**, accessible computer vision solutions and AI for physical robots, from idea to working product. Skilled in **Python, tool/function calling, RAG, LangChain/LangGraph, structured outputs and evaluation**. Committed to **responsible AI**, rigorous testing, professional ethics and full ownership of outcomes.

Education

Amrita Vishwa Vidyapeetham, Amrita School of AI *2023 – 2027*
B.Tech – Computer Science and Engineering (Artificial Intelligence) CGPA: **8.06/10**

Notable Achievements

Guinness World Record Holder | *Largest Electronic Keyboard Ensemble (440 participants)* *2017*
4-time Hackathon Finalist | *Civic Stack (IIT Delhi), AWS AI for Bharat, Technova, 2nd SmallAI*

Experience

Latlon Technologies Pvt. Ltd. *May 2026 – Jun 2026*
AI/ML Intern Coimbatore, India

- Co-developed an **autonomous, LLM-powered AI agent** environment (Agentic AI) that analyzes and extends codebases.
- Automated code modernization by pairing **AI code generation** with build, testing and validation steps.

NetKathir Technologies Pvt. Ltd. *Apr 2025 – May 2025*
AI Developer Intern Puducherry, India

- Built an **AI-powered 3D medical conversational avatar** with a **3-stage speech pipeline** (speech-to-text, NLP response generation, text-to-speech) for real-time patient queries.

Projects

Lazarus – Autonomous Agentic AI Platform for Legacy Code Modernization **Jan 2026 – Mar 2026**
Agentic AI, Multi-Agent Systems | Python, AWS Bedrock, Lambda, Step Functions, DynamoDB, S3, Docker

- Built a **5-agent autonomous AI pipeline** (Inspector, Architect, Builder, Deployer, Validator) on **AWS Bedrock** with a **self-healing agent loop** that reads CloudWatch logs and patches failures: **97% build success in about 8 minutes**.
- Agents perform **AST-based code analysis**, **dependency-graph extraction** and **automated code migration** of legacy repositories, with LLM-driven planning, code generation and validation on **AWS Bedrock**.
- Orchestrated the agents serverlessly on **AWS Lambda, Step Functions and API Gateway** with state in **DynamoDB** and artifacts in **S3**, deployed with **Docker**.

Continuous Sign Language Recognition for Deaf & Mute Users (GCN–TCN) **Oct 2025 – Apr 2026**
Computer Vision, Deep Learning, Accessibility | Python, PyTorch, MediaPipe, OpenCV, NumPy

- Built a **webcam-based sentence-level sign language recognition** pipeline (spatial **GCN** + dilated **TCN**) to make video calls accessible: **85.71% accuracy, 0.893 macro F1** on 10 ISL sentences, up from 50%.
- Introduced **eigenvalue-based joint selection** (35 joints, 78% of motion variance, 53% fewer nodes) to fix overfitting, plus **signer-adaptive joint weighting** at inference.

UrbanPulse – AI-Based Civic Issue Monitoring & Decision Support System **Dec 2025 – Mar 2026**
5-member team | Python, CNN, DistilBERT, Grad-CAM, Kalman Filter, React Native, Supabase

- Built the **AI verification layer**: a Roboflow **CNN** with **Grad-CAM** validates issue photos, **DistilBERT** scores urgency from text and voice input, and a scraper ingests issues from 3 social platforms, with spoofing checks.
- Designed **AHP-weighted priority scoring** over 4 signals (image, urgency, proximity, social) and a **Kalman-filter severity forecast**, surfaced as 3-km heatmaps and an admin **explainable AI (XAI)** dashboard.

Autonomous Plastic-Waste Collection Robot using Reinforcement Learning **May 2026 – Aug 2026**
Reinforcement Learning, Robotics | Python, Q-Learning, ROS 2, Gazebo, Nav2, RTAB-Map, RealSense

- Built a **resource-aware RL agent** that replaced if/else rules with **tabular Q-learning**: it picks **SEEK, PATROL or RETURN** from battery, distance to plastic and bin, and waste availability (**Bellman updates**, **ϵ-greedy**).
- Deployed in **Gazebo** and on a **real rover** (object detection, depth obstacle avoidance, RTAB-Map + Nav2, emergency stop): **26 of 30 bottles (87%) collected** across 10 real-world trials, full task success in 6.

Technical Skills

- **AI Agents & LLMs:** AI Agents, Multi-Agent Systems, Agentic Workflows, Tool/Function Calling, Model Context Protocol (MCP), RAG, LangChain, LangGraph, Prompt Engineering, Structured Outputs, LLM Evals, Guardrails, AWS Bedrock, Azure AI Foundry
- **ML & Vision:** Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Fine-Tuning (LoRA), PyTorch, TensorFlow, OpenCV, Hugging Face
- **Languages:** Python, SQL, Java, C++, JavaScript and TypeScript (basics)
- **Backend, Data & DevOps:** FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MongoDB, Vector Database, Docker, AWS, GCP, GitHub Actions (CI/CD), Git, Claude Code, Cursor

Certifications

- **Microsoft Applied Skills: Develop an Agent with Integrated Tools** | *Microsoft*
- **Hugging Face Agents Course – Certificate of Excellence** | *Hugging Face*
- AWS Certified: **AI Practitioner (AIF-C01)** and **Cloud Practitioner (CLF-C02)** | *Amazon Web Services*